Highlights

Characterisation and Quantification of Data Centre Flexibility for Power System Support

- An integrated model coordinates IT workloads, UPS batteries and cooling.
- Flexibility is quantified by feasible power deviations and service durations.
- Coordinated operation reduces annual electricity costs by 5.38%.
- A 1 MW IT-rated site sustains a 100 kW load reduction for six hours at noon.
- Workload flexibility has the largest savings effect among tested resources.